AP & RAP Assignment 3

1. Please check the correct statements below. Here, $(\Omega, \mathcal{A})$ is a measurable space and $f, g : \Omega \to \bar{\mathbb{R}}$ are functions.
   ☐ Suppose $\mathcal{A} := \{\emptyset, \Omega\}$. Then all measurable functions on $(\Omega, \mathcal{A})$ are constant functions.
   ☐ If $f + g$ is measurable, then f and g are both measurable.
   ☐ If f and g are measurable, then $\{\omega \in \Omega : f^2(\omega) \leq g(\omega)\} \in \mathcal{A}$.
   ☐ If f^2 is measurable, then f is also measurable.
   ☐ If f^3 is measurable, then f is also measurable.
   ☐ If f is measurable, then $\{f(B) : B \in \mathcal{A}\}$ is a σ-algebra on $\bar{\mathbb{R}}$.
   ☐ If f and g are measurable, then $f1_{\{g>c\}}$ is also measurable, where $c > 0$ is a constant.
   ☐ Suppose that for every $n \in \mathbb{N}$, $f1_{\{|f|\leq n\}}$ is measurable. Then f is also measurable.
   ☐ Suppose that for every $\alpha \in \mathbb{R}$, $f_\alpha$is measurable. Then $\sup_{\alpha\in\mathbb{R}} f_\alpha$ is also measurable.

2. Let f and g be extended real-valued measurable functions on $(\Omega, \mathcal{A})$, and define

$$h(\omega) = \begin{cases} f(\omega) & \text{if} \quad \omega \in A \\ g(\omega) & \text{if} \quad \omega \in A^c \end{cases}$$

   where A is a set in $\mathcal{A}$. Show that h is measurable.

3. Show that $f \in \mathcal{E}$ (set of all simple functions on a measurable space) implies that $f^\pm \in \mathcal{E}$. Is the converse valid?

4. Show that for every function $u : \Omega \to \bar{\mathbb{R}}$, it holds $u = u^+ - u^-$ and $|u| = u^+ + u^-$.